

Omar Eduardo Borges Montero

AI & Automation Engineering Student — Double Degree (Udine–Klagenfurt)

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Povoletto (UD), Italy



Profile

Master's student in **AI & Automation Engineering** with a strong international background (Double Degree Italy–Austria and two Erasmus in Spain). Highly motivated to apply frontier technologies in **Defense, Aerospace, Cybersecurity, and Motorsport**. Solid hands-on experience in Deep Learning, Computer Vision, embedded systems, and data pipelines. Competitive track-and-field athlete, bringing resilience and stress-management to dynamic, multicultural team environments. My goal is to design intelligent, secure, and responsible systems (Trustworthy AI).

Education

MSc in Multimedia Communication and Information Technologies (AI, Automation & IoT)

University of Udine (Italy) & University of Klagenfurt (Austria) — *Double Degree* (2024–2026)

Current weighted average: 29.143/30

- **Focus:** Machine Learning, Computer Vision, IoT, Embedded Systems, Cybersecurity, Intelligent Control Systems.
- **Double Degree (Austria):** Currently attending advanced courses at Uni Klagenfurt, including *Neurocomputing in Robotics & Intelligent Transportation*, *Sensor Networks* (Wasmote, XBee Pro), *Wireless Networks & WiFi Security*, *Sensors & Actuators* (Soft Robotics), *Artificial Vision*, and *Mobile Applications* (Android/iOS).
- **Erasmus+:** Universidad de Alcalá (Madrid). Focus on Data Science Fundamentals, Intelligent Control Systems, Artificial Intelligence.

BSc in Multimedia Sciences and Technologies — University of Udine (2021–2024)

Grade: 82/110

- **Focus:** Programming, Computer Science, Mathematics, and Interactive Media (AR/VR).
- **Erasmus+:** Universitat Politècnica de València (UPV) — Spain.
- **Thesis:** Technical and thematic analysis of *Interstellar* and its soundtrack.

High School Diploma in Electronics — ITI A. Malignani (2016–2021)

Grade: 74/100

Technical Projects

F1 Telemetry Analysis

- Developed a tool to extract and visualize real-time telemetry data from Formula 1 sessions.
- Analyzed lap times, speed profiles, and braking points to gain insights into vehicle performance.
- Tools: Python, FastF1, Pandas, Matplotlib.

Real-Time Object Detection (YOLOv8)

- Implemented a computer vision system for real-time detection and tracking using the YOLOv8 architecture.
- Focused on model optimization for low-latency inference on custom datasets.
- Tools: Python, Ultralytics, OpenCV.

SecureCV: Cryptographic Document Protection

- Designed a security layer for document integrity and confidentiality using encryption and hashing algorithms.
- Tools: Python, Cryptography.

HuBMAP + HPA: Biomedical Image Segmentation (Kaggle)

- Developed a segmentation pipeline using U-Net and ensemble architectures for functional tissue units.
- Tools: PyTorch, MONAI, OpenCV.

Neural Network Position Control

- Designed and tested a feedforward neural network for dynamic trajectory learning in MATLAB/Simulink.
- Tools: MATLAB, Simulink.

Recent Trends in IoT Sensor Technologies — Research Seminar

- Writing a comprehensive research paper (25+ pages) on pervasive computing and IoT sensor networks applied to Smart Agriculture.
- Analyzing state-of-the-art wireless sensor nodes and data architectures for precision farming.

DocuStudent: Mobile Document Archive — Mobile Applications Course

- Developing an Android/iOS app for fast document scanning, categorization, and digital archiving for university students.
- Implementing local/cloud storage integration, intuitive UI/UX, and search filtering capabilities.
- Tools: Java, Android Studio.

Skills

Programming: Python, C/C++, MATLAB, Java, R, SQL, JavaScript, HTML5/CSS3

AI/ML: PyTorch, TensorFlow, scikit-learn, YOLOv8, MONAI, OpenCV

Concepts: CNNs, U-Net, Object Detection, Cryptography, WiFi Security, ML pipelines

Tools: Git, Android Studio, LaTeX, WordPress

Embedded/IoT: Wasmote, XBee Pro, Microcontrollers, Bluetooth/Wireless protocols

Experience

Web Developer Intern — AdaLab Udine (Mar–Apr 2024)

- Managed full-cycle development of responsive websites using HTML5, CSS3, JavaScript, and WordPress.
- Gained strong skills in independent project management and structured team collaboration.

Waiter Assistant — Adria Sole S.N.C. (Summer 2021, 2022)

- Handled high-pressure shifts in a fast-paced environment, excelling in stress management.
- Developed strong operational problem-solving and cross-team communication skills.

Certifications & Advanced Training

Advanced Training Course: "IA, quo vadis?" — EFC Network (Feb 2026)

Focus: AI Ethics, European Regulatory Framework (AI Act), Privacy, and Trustworthy AI.

ENAC UAS A1–A3 Drone Pilot License

Google Digital Marketing (Fundamentals)

Workshop: Digital Innovation and AI in Mountain Environments

Languages

Italian (Native), Spanish (Native), English (B2)

Extracurricular

UEFA Super Cup 2025 Volunteer — Ceremony Maker

Active runner; participation in athletics events.

Interests

AI, defense technologies, motorsport, running, innovation